# INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

## THE SITE

| | |
|---|---|
| Location | QTS Manassas DC5, VA -- station KHEF, America/New_York |
| Committed pair | OSM 1287260559 -> 1181391922<br>OSM way 1287260559 / Amazon IAD131 |
| Facade gap | 131.1 m between the two halls |
| Weather record | 42,554 real hourly records from KHEF |
| Plume physics | 576 steady-state solves on this site's own footprints; worst intake rise 0.4115 C at 165 deg |

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

## THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

| | |
|---|---|
| Day | 2023-04-11 (crossing) |
| Plant limit | 18.0 C |
| Notice required | 3 h |
| Forecast skill | 0.50 relative to persistence |
| Level anchor | sensor |
| Condenser bank | longest facade |
| Switch budget | 1 mode changes per day |
| Minimum dwell | 3 h |
| Max dew point | 15.0 C (Green Grid WP#46 p.6) |

## WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 3 of 24 hours with 1 mode change. The reactive incumbent took 12 hours -- 9 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 11 hours were safe but still ran chillers, because the schedule could not afford them.

| | |
|---|---|
| Agent | 3 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s) |
| Incumbent | 12 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)<br>the incumbent broke its own switch budget 1 time(s) to stay safe<br>11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them |

## EVERY HOUR, AND THE REASON FOR IT

| hh | mode | safe | binding | bound | limit | actual | margin |
|---|---|---|---|---|---|---|---|
| 00 | mechanical | yes | switch budget | 5.377 | 18.0 | 5.043 | 1.135 |
| 01 | mechanical | yes | switch budget | 5.273 | 18.0 | 4.104 | 1.343 |
| 02 | mechanical | yes | switch budget | 4.443 | 18.0 | 2.433 | 1.240 |
| 03 | mechanical | yes | switch budget | 4.723 | 18.0 | 2.994 | 1.198 |
| 04 | mechanical | yes | switch budget | 3.333 | 18.0 | 1.323 | 1.144 |
| 05 | mechanical | yes | switch budget | 1.968 | 18.0 | 0.213 | 1.111 |
| 06 | mechanical | yes | switch budget | 2.468 | 18.0 | -0.397 | 1.416 |
| 07 | mechanical | yes | switch budget | 4.998 | 18.0 | 4.104 | 1.722 |
| 08 | mechanical | yes | switch budget | 8.888 | 18.0 | 9.104 | 2.451 |
| 09 | mechanical | yes | switch budget | 11.932 | 18.0 | 12.868 | 2.859 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 10 | mechanical | yes | switch budget | 16.388 | 18.0 | 16.884 | 3.007 |
| 11 | mechanical | no | dry-bulb | 19.317 | 18.0 | 20.109 | 2.299 |
| 12 | mechanical | no | dry-bulb | 21.668 | 18.0 | 22.994 | 1.837 |
| 13 | mechanical | no | dry-bulb | 23.474 | 18.0 | 24.244 | 1.503 |
| 14 | mechanical | no | dry-bulb | 24.904 | 18.0 | 24.839 | 1.389 |
| 15 | mechanical | no | dry-bulb | 26.217 | 18.0 | 25.560 | 1.284 |
| 16 | mechanical | no | dry-bulb | 25.516 | 18.0 | 25.000 | 1.260 |
| 17 | mechanical | no | dry-bulb | 25.069 | 18.0 | 24.446 | 1.473 |
| 18 | mechanical | no | dry-bulb | 24.444 | 18.0 | 22.805 | 1.741 |
| 19 | mechanical | no | dry-bulb | 21.938 | 18.0 | 18.544 | 1.953 |
| 20 | mechanical | no | dry-bulb | 20.278 | 18.0 | 15.774 | 2.066 |
| 21 | FREE | yes | - | 17.778 | 18.0 | 12.434 | 1.915 |
| 22 | FREE | yes | - | 15.273 | 18.0 | 11.323 | 1.796 |
| 23 | FREE | yes | - | 12.498 | 18.0 | 8.543 | 1.449 |

bound  = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

## THE REASONING, HOUR BY HOUR

**00:00  MECHANICAL  [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.377 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**01:00  MECHANICAL  [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.273 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**02:00  MECHANICAL  [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.443 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**03:00  MECHANICAL  [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.723 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**04:00  MECHANICAL  [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.333 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**05:00  MECHANICAL  [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.968 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**06:00  MECHANICAL  [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.468 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**07:00  MECHANICAL  [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.998 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**08:00  MECHANICAL  [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.888 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**09:00  MECHANICAL  [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.932 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**10:00  MECHANICAL  [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.388 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**11:00  MECHANICAL  [dry-bulb]**

Mechanical. The upper bound on intake air is 19.317 C against a 18.0 C limit, so it fails by 1.317 C. It would take a limit 1.317 C higher, or a bound 1.317 C tighter, to change this hour.

**12:00  MECHANICAL  [dry-bulb]**

Mechanical. The upper bound on intake air is 21.668 C against a 18.0 C limit, so it fails by 3.668 C. It would take a limit 3.668 C higher, or a bound 3.668 C tighter, to change this hour.

**13:00  MECHANICAL  [dry-bulb]**

Mechanical. The upper bound on intake air is 23.474 C against a 18.0 C limit, so it fails by 5.474 C. It would take a limit 5.474 C higher, or a bound 5.474 C tighter, to change this hour.

**14:00  MECHANICAL  [dry-bulb]**

Mechanical. The upper bound on intake air is 24.904 C against a 18.0 C limit, so it fails by 6.904 C. It would take a limit 6.904 C higher, or a bound 6.904 C tighter, to change this hour.

**15:00  MECHANICAL  [dry-bulb]**

Mechanical. The upper bound on intake air is 26.217 C against a 18.0 C limit, so it fails by 8.217 C. It would take a limit 8.217 C higher, or a bound 8.217 C tighter, to change this hour.

**16:00  MECHANICAL  [dry-bulb]**

Mechanical. The upper bound on intake air is 25.516 C against a 18.0 C limit, so it fails by 7.516 C. It would take a limit 7.516 C higher, or a bound 7.516 C tighter, to change this hour.

**17:00  MECHANICAL  [dry-bulb]**

Mechanical. The upper bound on intake air is 25.069 C against a 18.0 C limit, so it fails by 7.069 C. It would take a limit 7.069 C higher, or a bound 7.069 C tighter, to change this hour.

**18:00  MECHANICAL  [dry-bulb]**

Mechanical. The upper bound on intake air is 24.443 C against a 18.0 C limit, so it fails by 6.443 C. It would take a limit 6.443 C higher, or a bound 6.443 C tighter, to change this hour.

**19:00  MECHANICAL  [dry-bulb]**

Mechanical. The upper bound on intake air is 21.938 C against a 18.0 C limit, so it fails by 3.938 C. It would take a limit 3.938 C higher, or a bound 3.938 C tighter, to change this hour.

**20:00  MECHANICAL  [dry-bulb]**

Mechanical. The upper bound on intake air is 20.278 C against a 18.0 C limit, so it fails by 2.278 C. It would take a limit 2.278 C higher, or a bound 2.278 C tighter, to change this hour.

**21:00  FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 17.778 C, which is 0.222 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.915 C, and the margin is measured, not chosen: 1.575 C of group-conditional forecast error for this hour of day, plus 0.3399 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**22:00  FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 15.273 C, which is 2.727 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.796 C, and the margin is measured, not chosen: 1.456 C of group-conditional forecast error for this hour of day, plus 0.3399 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**23:00  FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 12.498 C, which is 5.502 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.449 C, and the margin is measured, not chosen: 1.109 C of group-conditional forecast error for this hour of day, plus 0.3399 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

Generated by INTAKE-ARBITER/src/report.py. Verified by being read back: the file was reopened after writing and every hour of the schedule, the headline counts and this site's own name were confirmed present.